

Department of Electrical and Electronics Engineering

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Title: Digital Electronics

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SYLLABUS

Introductory Concepts: Digital and Analog Quantities, Binary Digits, Logic Levels, Digital Electronic Signals and Switches, Basic Logic Functions, Combinational and Sequential Logic Functions, Introduction to Programmable Logic and Fixed-Function Logic Devices.

Number Systems, Operations, and Codes: Decimal, Binary, Hexadecimal, Octal and Binary Coded Decimal Number Systems, Binary Arithmetic and Digital Codes.

Logic Gates: Inverter, AND Gate, OR Gate, NAND Gate, NOR Gate, ExclusiveOR and Exclusive-NOR Gate, Universal Gate, Programmable Logic, FixedFunction Logic Gates.

Boolean Algebra and Logic Simplification: Boolean Operations, Expressions,Laws and Algebra, De Morgan's Theorems, Boolean Analysis of Logic Circuits, The Karnaugh Map, Don't Care Conditions, SOP and POS Minimization.

Arithmetic Operations and Circuits: Half and Full Adders, Half and Full Subtractor, 2's complement, Ripple Carry and Look-Ahead Carry Adders, BCD Adders, Cascading BCD Adders, Multipliers.

Functions of Combinational Logic: Decoders: 1-of-8 decoders, BCD-to-Decimal Decoder/Driver, BCD-to-7-Segment Decoder/Drivers, Decoder IC, Encoders and Applications, Priority Encoders, Code Converters, Magnitude Comparators, Parity.Generators/Checkers, Multiplexers: Two, Four, Eight, Sixteen, Quad Two-Input MUX, Multiplexer Applications: Data Routing, Parallel-to-Serial Conversion, Logic Functions Generation, Demultiplexers and its Applications, 1-Line-to-8-Line Demultiplexer, Synchronous Data Transmission.

Sequential Logic: NAND/NOR Gate Latch, Edge-Detector Circuit, Flip-Flops: Clocked S-C Flip-Flop, Clocked J-K Flip-Flop, Clocked D Flip-Flop, T Flip-Flop, Master/Slave Flip-Flop, Clocked J-K Flip-Flop with Asynchronous Inputs, FlipFlop Applications: Edge Detection, Switch Bouncing Reduction, Parallel Data Transfer, Serial Data Transfer. Shift Registers and Counters: Shift Register Operations, Types of Shift Register, Shift Register Counters: Ring Counter and Johnson Counter, Asynchronous Counters, IC Asynchronous Counters, Asynchronous Down Counters, Synchronous Up/Down Counters, Pre-settable Counters, Counter with Arbitrary Sequences, Design of Synchronous Counters and Cascaded Counters.

Interfacing with the Analog World:Methods of Digital-to-Analog Conversion: Weighted Resistors Converter, R-2R Ladder Converter, Methods of Analog-toDigital Conversion: Flash, Digital Ramp, Successive Approximation Converter, ADC and DAC Applications.

Integrated-Circuit Logic Families: Characteristic Parameters, The TTL Logic Family, TTL Series Characteristics, TTL Circuit Operation, Current-Sinking and Sourcing Action,

Totem-Pole Output Circuit, TTL Series Characteristics, TTL Loading, Fan-In and Fan-Out, TTL NOR Gate, TTL NAND Gate, Digital MOSFET Circuits: CMOS, NMOS and PMOS Gates, Emitter-Coupled Logic, Direct Coupled Transistor Logic, Resistor Transistor Logic.

Finite State Machines: State Diagram, Moore-type and Melay-type Machines, State Machines Synthesis, State Machines in Verilog, State Encoding.

Introduction to Programmable Logic Device: Programmable Logic Device (PLD), Programmable Logic Array (PLA), Field-Programmable Gate Array (FPGA).

REFERENCE BOOKS

1. Digital Systems, 12th Edition, Ronald Tocci, Neal Widmer and Greg Moss, Pearson.
2. Digital Fundamentals, 11th Edition, Thomas L. Floyd, Pearson.
3. Digital Electronics: A Practical Approach, 8th Edition, William Kleitz, Prentice Hall.
4. Digital Electronics: Principles, Devices and Applications, Anil K. Maini, John Wiley and Sons Pub. Ltd.

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CHAPTER 1

INTRODUCTORY CONCEPTS

1.1 Signal

Signal : A signal is a function that represents the variation of a physical quantity with respect to any parameter.

Here parameters are **time and distance** which are independent quantity.

→ In electrical and electronics, usually signal is variation of electrical quantity (Generally I and V) with respect to time.

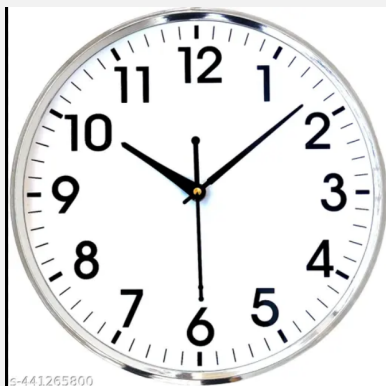
If the current I or voltage V is remain same with respect to t , it's not a signal it's a direct current. Signal must derived with the independent quantity.

■ **Transducer :** Its a device which is used to convert non-electrical signal to an electrical signal.

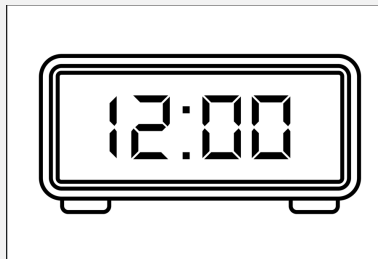
■ **Reverse Transducer :** Convert electrical signal to non-electrical signal.

Analog Signal : An analog signal is a type of signal that varies continuously over time and can take on any value within a given range.

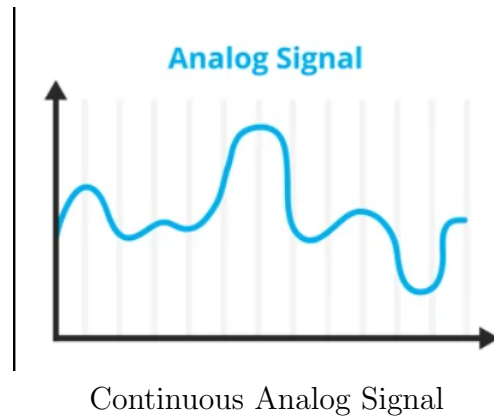
Q 1.0: Why Clock 1 is called analog and clock 2 is called digital?



we can have every value in the given limit.



In this clock we can't have 12:00:30 or 12:00:15.



We can have every value in the given range. So this function is called *Analog signal*.

Discrete Time Signal : The signal which is defined for the discrete interval of time is called Discrete time signal.

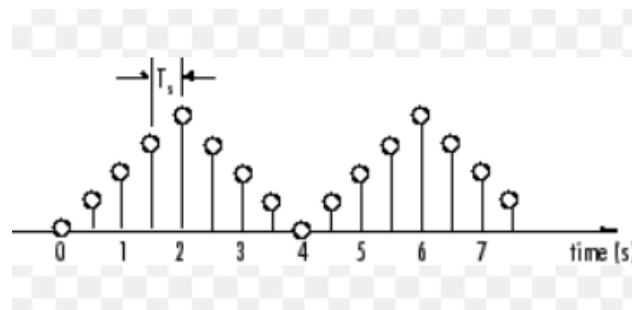


Figure 1.1: Discrete time signal

From this graph we have the value at $t_0, t_1, t_2, \dots, t_n$, but we can't have any value between t_0 and t_1 . Because we have monitored the function for the particular time.

- * Discrete time signal is the subset of analog signal.
- * All real life signal is analog signal.

Example: Sound, Temperature, Speed are analog signal.

Digital Signal : A digital signal is a type of signal that represents data using discrete values, typically in the form of binary numbers (0s and 1s). Unlike analog signals which are continuous, digital signals jump between fixed levels.

→ In digital signals, we discretize both time and magnitude.

→ We divide the magnitude axis into a fixed number of levels and the signal can take values equal to these levels only.

For digital signal we have to make a level to minimize the error. **Increase the no of level reduce the error.**

Again to minimize the value we have to take the lower value of the level. At t_1 , T is 9°C . It

is closer to $15^{\circ}C$ but we can't take this value. We have to take $0^{\circ}C$. **The keypoint is we don't select the higher values ,we can just select the lower values.**

Q 1.1: What is the need of Digital Signal?

→ Digital signal is used in communication process to minimize the effect of noise (unwanted signals).

If the Analog signal voltage is $[0 - 5] V$ and we want to transmit this into $2.5 V$.

But from this graph, there is no $2.7V$ level. So we can't take the voltage, we should take the lower level that is $2.5V$. This is the actual signal. This way Digital signal minimizes the noise.

1.2 Introduction of Digital Electronics

Digital System → Sub system → Modules → Basic Units (Logic gate)
→ Circuit → Transistor, Capacitor, Resistor.

Switch and Bits intuition

■ Advantage of Digital System :

- (i) Noise immunity.
- (ii) Uses less bandwidth.
- (iii) Encryption.
- (iv) Efficiency for long distance transmission.
- (v) Accuracy.

* If we want to increase the accuracy, we have to increase no. of level.

* $n =$ no of levels
 $m =$ no of switch

$$n = 2^m$$

For 3 switch we have to need 8 levels for increasing accuracy.

In 2 bits number we have 4 possibility.

*** $\boxed{\text{no of bits} = \text{no of switch}}$

1.3 Introduction to Boolean Algebra

→ It is the set of rules used to simplify the given logic expression without changing its functionality.

→ It is used when numbers of variables are less (like 1, 2, 3).

→ For more variables (like 4, 5, 6) we use another method which is known as **Map Methods**, also known as **Karnaugh Map or K-Map**.

- Why do we minimize expression? → To minimize gates.

Q 1.2: What is Logic expression?

Rules of Boolean algebra :

1. **Complement** : A complement = $\bar{A}$ or A' (not A)

$$* (A')' = A \quad [0' = 1; 1' = 0]$$

2. **AND** :

- $A.A=A$
- $A.0=0$
- $A.1=A$
- $A.A'=0$

3. **OR** :

- $A+A=A$
- $A+0=A$
- $A+1=1$
- $A+A'=1$

4. **Distributive Law** :

- $A.(B+C)=AB+AC$
- $A+(B.C)=(A+B).(A+C)$
- $A+\bar{A}B=A+B$

5. **Commutative Law** :

- $A+B=B+A$
- $A.B=B.A$

6. **Associative Law** :

- $(A.B).C=A.(B.C)$

Priority : NOT → AND → OR.

Boolean Algebra Example :

Example 1.0: Solve $AB+AB'$

Let

$$\begin{aligned}
 F &= AB + AB' \\
 &= A(B + B') \\
 &= A
 \end{aligned}$$

Example 1.1: Solve $AB + A\bar{B}C + A\bar{B}\bar{C}$

Let

$$\begin{aligned}
 F &= AB + A\bar{B}C + A\bar{B}\bar{C} \\
 &= A[B + \bar{B}C + \bar{B}\bar{C}] \\
 &= A[B + \bar{B}(C + \bar{C})] \\
 &= A[B + \bar{B}] = A
 \end{aligned}$$

Example 1.2: Solve $(A + B + C)(A + \bar{B} + C)(A + B + \bar{C})$ Let, $X=A+B$

$$\begin{aligned}
 F &= (A + B + C)(A + \bar{B} + C)(A + B + \bar{C}) \\
 &= (X + C)(A + \bar{B} + C)(X + \bar{C}) \\
 &= (X + C)(X + \bar{C})(A + \bar{B} + C) \\
 &= (X + C\bar{C})(A + \bar{B} + C) \\
 &= (X + 0)(A + \bar{B} + C) \\
 &= (A + B)(A + \bar{B} + C) \\
 &= A + B(\bar{B} + C) \\
 &= A + BC = (A + B)(A + C) \\
 &= A + B\bar{B} + BC = A + BC
 \end{aligned}$$

1.4 Redundancy Theorem :

1. Three variable.
2. Each variable is repeated twice.
3. One variable is complimented
4. Take the compliment variable.

Example 1.3: Simplify $Y = AB + \bar{A}\bar{C} + BC$

$$\begin{aligned}
 Y &= AB + \bar{A}\bar{C} + BC \\
 &= AB + \bar{A}\bar{C} + BC.1 \\
 &= AB + \bar{A}\bar{C} + BC(A + \bar{A}) \\
 &= AB + \bar{A}\bar{C} + \bar{A}BC + ABC \\
 &= AB(1 + C) + \bar{A}\bar{C}(1 + B) \\
 &= AB + \bar{A}\bar{C}
 \end{aligned}$$

Example 1.4: Simplify $F = AB + \bar{B}\bar{C} + AC$

$$\begin{aligned}
 F &= AB + \bar{B}\bar{C} + AC \\
 &= AB + \bar{B}\bar{C} + AC.1 \\
 &= AB + \bar{B}\bar{C} + AC(C + \bar{C}) \\
 &= AB + ABC + \bar{B}\bar{C} + A\bar{B}\bar{C} \\
 &= AB(1 + C) + \bar{B}\bar{C}(1 + A) \\
 &= AB + \bar{B}\bar{C}
 \end{aligned}$$

Short Cut: The answer is the sum of component without complement element and the component with complement element.

Example 1.5: Simplify

1. $Y = A\bar{B} + BC + AC$

2. $F = \bar{A}\bar{B} + A\bar{C} + \bar{B}\bar{C}$

$$\begin{aligned}
 1.Y &= A\bar{B} + BC + AC \\
 &= AC + A\bar{B}
 \end{aligned}$$

$$\begin{aligned}
 2.F &= \bar{A}\bar{B} + A\bar{C} + \bar{B}\bar{C} \\
 &= \bar{A}\bar{B} + A\bar{C}
 \end{aligned}$$

Example 1.6: Simplify

1. $F = (A + B)(A + C)(\bar{B} + C)$

2. $F = (A + B) + (\bar{A} + C) + (B + C)$

$$\begin{aligned}
 1. F &= (A + B)(A + C)(\bar{B} + C) \\
 &= (A + B)(\bar{B} + C)
 \end{aligned}$$

$$\begin{aligned}
 2. F &= (A + B) + (\bar{A} + C) + (B + C) \\
 &= (A + B)(\bar{A} + C)
 \end{aligned}$$

1.5 Sum of Product (SOP)

SOP form only use for high function means o/p is 1.

Total no of Combination = 2^n [n → no of variables]

For 3 variables total combination = $2^3 = 8$

A	B	F
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

SOP form of this table :

$$\begin{aligned}
 F &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC \\
 &= \sum m(2, 4, 5, 6, 7)
 \end{aligned}$$

This particular form are called as standard or canonical SOP form and this is written directly from truth table.

Minterm → $\bar{A}BC, ABC, AB, BC$ etc.

Maxterm → $(A + B + C)$

Example 1.7: Write the minimal SOP form of this function
 $F = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$

$$\begin{aligned}
 F &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC \\
 &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}(\bar{C} + C) + AB(\bar{C} + C) \\
 &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B} + AB \\
 &= \bar{A}\bar{B}\bar{C} + A(\bar{B} + B) \\
 &= \bar{A}\bar{B}\bar{C} + A \\
 &= A + \bar{B}\bar{C} \quad [\because A + \bar{A}B = A + B]
 \end{aligned}$$

This is the minimal SOP form of this cononical SOP form.

There are two types of SOP form.

1. standard/Cononical : Each minterm is having all the variables in normal or complicated form.

Example : $\bar{A}B + AB + \bar{A}\bar{B}$

2. Minimal : Each minterm doesn't have all the variables in normal or complicated form.

Q 1.3: For the given truth table minimize the SOP expression.

A	B	F
0	0	0
0	0	1
0	1	0
0	1	1

The cononical SOP form of this function

$$\begin{aligned}
 F &= \bar{A}B + AB \\
 &= B(\bar{A} + A) \\
 &= B
 \end{aligned}$$

Q 1.4: Simplify the expression for $Y = (A + B) = \sum(0, 2, 3)$

$$\begin{aligned}
 Y &= M_0 + M_2 + M_3 \\
 &= \bar{A}\bar{B} + A\bar{B} + AB \\
 &= \bar{B}(\bar{A} + A) + AB \\
 &= \bar{B} + AB \\
 &= A + \bar{B}
 \end{aligned}$$

1.6 Product of Sum (POS form)

→ POS form is used for low function means the O/P is '0'.

$0 \rightarrow A$
 $1 \rightarrow \bar{A}$
 POS form

$0 \rightarrow \bar{A}$
 $1 \rightarrow A$
 SOP form

Example 1.8: Write the minimize POS form of this truth table.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$\begin{aligned}
 F &= (A + B + C).(A + B + \bar{C})(A + \bar{B} + \bar{C}) \rightarrow [\text{maxterm}] \\
 &= (X + C)(X + \bar{C})(A + \bar{B} + \bar{C}) \\
 &= (X + C\bar{C})(A + \bar{B} + \bar{C}) \\
 &= A + B(\bar{B} + \bar{C}) \\
 &= A + B\bar{B} + B\bar{C} \\
 &= A + B\bar{C} \\
 &= (A + B)(A + C) \rightarrow [\text{Minimal POS form}]
 \end{aligned}$$

Q 1.5: For the given truth table minimize the POS form

A	B	Y
0	0	0
0	0	1
0	1	0
0	1	1

$$\begin{aligned}
 Y &= (A + \bar{B})(\bar{A} + \bar{B}) \\
 &= \bar{B} + A\bar{A} \\
 &= \bar{B}
 \end{aligned}$$

$$\boxed{\prod (M_1, M_2)} = \prod_{\text{POS}} M(1, 3) = \sum_{\text{SOP}} M(0, 2)$$

Important

- SOP form → AND [more] → [If the equation has many AND gate, we use SOP]
- POS form → OR [more] → [If the equation has many OR gate, we use POS]

Example of some SOP and POS form

For those type of question, the truth table and equation are given.

Example 1.9: Minimize the equation

$$Y(A, B, C) = \sum M(0, 2, 3, 6, 7)$$

$$Y(A, B, C) = \prod M(1, 4, 5)$$

OR

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

POS

SOP

$$\begin{aligned}
 Y &= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C} + ABC \\
 &= \bar{A}\bar{B}\bar{C} + \bar{A}B(\bar{C} + C) + AB(\bar{C} + C) \\
 &= \bar{A}\bar{B}\bar{C} + \bar{A}B + AB \\
 &= \bar{A}\bar{B}\bar{C} + B(\bar{A} + A) \\
 &= \bar{A}\bar{B}\bar{C} + B \\
 &= B + \bar{A}\bar{C}
 \end{aligned}$$

$$\begin{aligned}
 Y &= (A + B + \bar{C})(\bar{A} + B + C)(\bar{A} + B + \bar{C}) \\
 &= (A + B + \bar{C})(\bar{A} + B + C\bar{C}) \\
 &= (A + B + \bar{C})(\bar{A} + B) \\
 &= B + (A + \bar{C})\bar{A} \\
 &= B + \bar{A}\bar{C}
 \end{aligned}$$

1.7 Minimal to Cononical form Conversion

For SOP form,

Step-1 Look at the variables and what the number of variables.

Step-2 Find missing variables.

Step-3 In the term of missing variable we put this variable and its complement.

Step-4 Ignore repeated part.

For POS form,

Step-1 Look at the variables and what the number of variables.

Step-2 Find missing variables.

Step-3 Retrieve the absent variable.

Example 1.10: Convert this in cononical form $Y = A + \bar{B}C$

$$\begin{aligned}
 Y &= A + \bar{B}C \\
 &= A.1.1 + 1.\bar{B}C \\
 &= A(B + \bar{B})(C + \bar{C}) + (A + \bar{A})\bar{B}C \\
 &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C \\
 &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C
 \end{aligned}$$

$$\begin{aligned}
 Y &= (A + B + \bar{C})(\bar{A} + C) \\
 &= (A + B + \bar{C})(\bar{A} + C + 0) \\
 &= (A + B + \bar{C})(\bar{A} + C + B\bar{B}) \\
 &= (A + B + \bar{C})(X + B\bar{B}) \\
 &= (A + B + \bar{C})(X + B)(X + \bar{B}) \\
 &= (A + B + \bar{C})(\bar{A} + B + C)(\bar{A} + \bar{B} + C)
 \end{aligned}$$

Q 1.6: In a minimal SOP form, how many no of minterms in the logical expression $A + \bar{B}C$

$$\begin{aligned}
 F &= A + \bar{B}C \\
 &= A.1.1 + 1.\bar{B}C \\
 &= A(B + \bar{B})(C + \bar{C}) + (A + \bar{A})\bar{B}C \\
 &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C \\
 &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C
 \end{aligned}$$

No of term=5

Q 1.7: With two variables, how many no of maximum minterms and maxterms?

$A, B \longrightarrow 2^2 = 4 \longrightarrow$ Possible combination.
 For n variable $\longrightarrow 2^n =$ Maxterms or Minterms.

Q 1.8: For $n = 4$, what is the total number of logical expression?

For $n=2$,